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## Topic 6 Test

ATTEMPT SCORE

40 / 50

### 01 Multiple Choice

2 / 2

**Choose the best answer.**

Low pressure systems often bring what kind of weather?

- ☐ cold and snowy weather
- ☐ warm and dry weather
- ☐ clear and sunny weather
- ☒ rainy or stormy weather



### 02 Matching

10 / 10

**Matching.**

Match the example to the correct vocabulary term. *Not all examples will be used!*

precipitation

A any form of condensed water falling from a cloud ✓

hail

B ice crystals that grow bigger in size as they get juggled up through a cloud -- and can really hurt if they fall on you! ✓

jet stream

C high-speed winds at the top of the troposphere ✓

meteorologist

D a scientist who studies the weather ✓

cyclone

E a low-pressure system that, if formed over a warm ocean, can turn into a hurricane ✓

❖ a device made of a wet bulb and a dry bulb thermometer

❖ when a warm front is sandwiched by two cold fronts

**03 Multiple Choice****2 / 2**

**Select all that apply.**

Knowing that a meteorologist is a scientist whose job requires special training, Madison interviewed a local meteorologist for a science assignment. She wanted to find out what the important job duties of a meteorologist are. Choose the two that apply.

- ✓
- ☒ collect and analyze data to make predictions about the weather ✓
  - ☐ guess what the weather will be and report it on television
  - ☐ carry out investigations of insects and how they affect the environment
  - ☒ use observations and technology to determine how global patterns affect weather ✓

**04 Multiple Choice**

2 / 2

**Select all that apply.**

Alysa learned that an air mass is a very large body of air with similar temperatures, humidity, and pressure, and that air masses are constantly in motion. She knows the air masses change, depending on the temperature and moisture content of the region where they form. She looked up more information about what makes them move. What are the major causes for moving air masses in North America? Choose the two that apply.

- ☐ changing humidity
- ☐ low temperatures
- ☒ jet stream ✓
- ☒ prevailing westerlies ✓

**05 Multiple Choice**

2 / 2

**Choose the best answer.**

Amber lives in the mountains of Colorado. Her aunt came to visit and had difficulty breathing for a few days until she adjusted to the higher altitude. Which explanation best describes the reason for this difficulty?

- ☒ Decreased air pressure causes the density of air to decrease at higher altitudes. ✓
- ☐ Decreased air pressure causes the density of air to increase at higher altitudes.
- ☐ Increased air pressure causes the density of air to increase at higher altitudes.
- ☐ Increased air pressure causes the density of air to decrease at higher altitudes.

06 Multiple Choice

2 / 2

**Choose the best answer.**

Evan knows that the water cycle follows certain steps in a repeating cycle. When the sun heats water molecules, they increase speed and collide. On Sunday he noticed that a puddle on the sidewalk was getting smaller over time. Which step in the water cycle would Evan predict is occurring?

- ☐ precipitation
- ☒ evaporation
- ☐ condensation
- ☐ crystallization

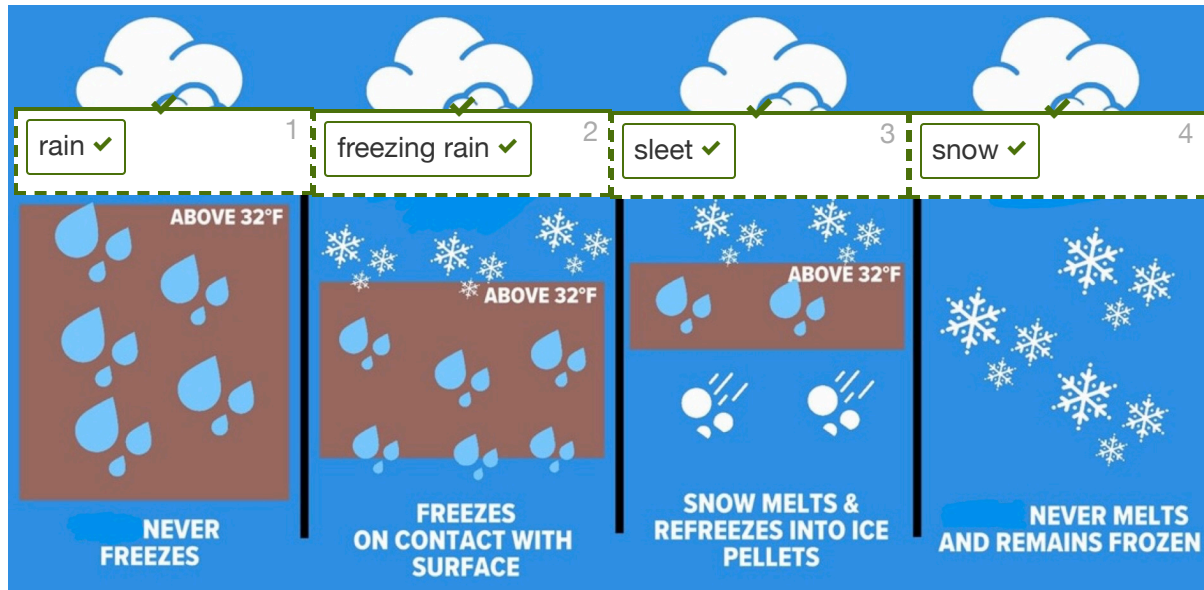


**07** Label Image

**2 / 2**

**Matching.**

Match the precipitation to the conditions that create it. *Not all types will be used!*



❏ hail

**Given responses:**

1 rain ✓

2 freezing rain ✓

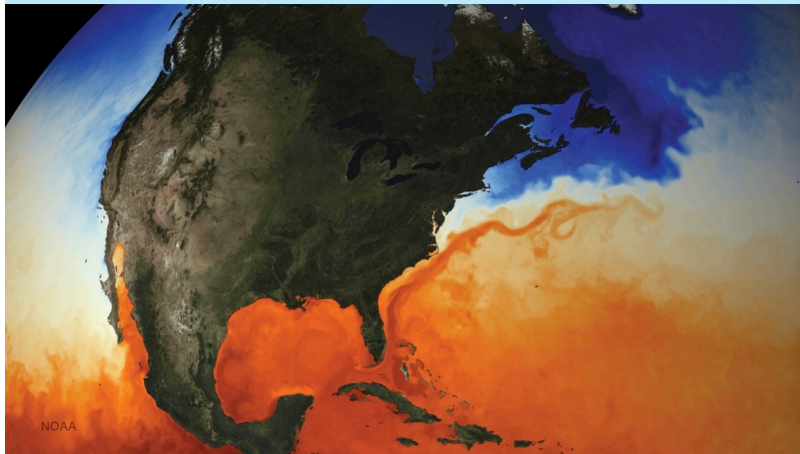
3 sleet ✓

4 snow ✓

## 08 True/False

1 / 2

**Ocean Surface Temperatures Figure 3** This satellite image uses red color to show the warm waters of the Gulf Stream ocean current as it moves north along the East coast into the cooler waters of the North Atlantic.



True or false.

The gulf stream brings a maritime polar air mass to the east coast, resulting in warmer temperatures.

☐ True

☒ False

**Student's Correction**

The gulf stream brings a maritime air mass to the east coast resulting in warmer temperatures.

**Correction Guidelines**

## 09 Multiple Choice

0 / 2

**Choose the best answer.**

Which of the following best explains what a high-pressure system does?

- ☒ sends out gusty winds in all directions
- ☐ pushes up warmer air which cools down and forms clouds
- ☐ pushes air down to the surface and prevents cloud formation
- ☐ creates unstable air conditions and thunderstorms

✗

✓

## 10 Multiple Choice

0 / 2

**Choose the best answer.**

Arizona often has hot, dry weather. What kind of air mass would most likely bring this sort of weather?

- ☐ maritime tropical
- ☐ continental tropical
- ☐ maritime polar
- ☒ continental polar

✓

✗

## 11 True/False

2 / 2



**True or false.**

On a cold February night, the dew point is 28°F. When water vapor condenses on your mom's Tesla overnight, it condenses as frost.

☒ True☐ False**12** Short Answer/Essay**3 / 6**

**Short answer.**

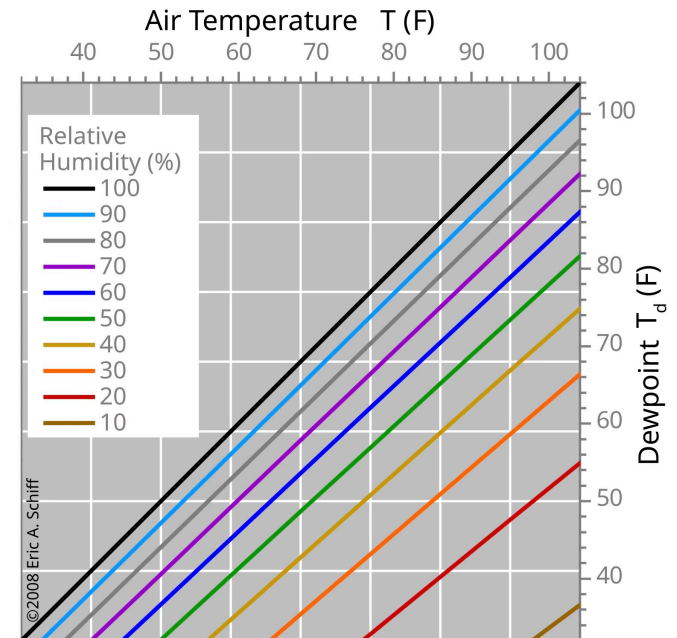
7:00 AM

**46°F**90% Humidity  
Dew Point: 43°

10:00 AM

**68°F**40% Humidity  
Dew Point: 43°

A weather station takes readings from the same location every hour. In a well-written paragraph, explain why the relative humidity changed between 7 AM and 10 AM, but the dew point didn't. Use information from the graph provided to support your answer.



The dew point didn't change because it is the temperature at which water condenses and relative humidity is the amount of water vapor currently compared to the maximum amount of water vapor. So in this case, the amount of water vapor changed but the rate didn't. At 7:00 AM, the relative humidity was 90%, so 90% of the air was water vapor. But that doesn't change the rate of water vapor condensing. At 10:00 AM, the temperature increased so the water vapor decreased, that is why at 10:00 AM the relative humidity decreased. But still, the rate at which the water vapor condenses didn't change. That is why the dew point didn't change as the relative humidity did.

118 / 500 Word Limit

**13 Matching****2 / 2**

**Matching.**

Kihara wanted to find out more about how technology helps to predict the weather. She found information and some pictures. For each picture below, choose a word from the word bank to complete the label naming each type of technology that meteorologists use to collect data and predict the weather.



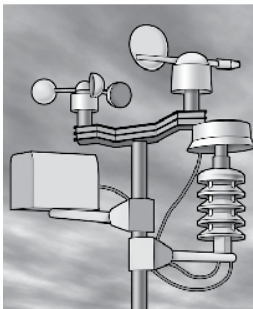
A

weather satellite



B

weather balloon



C

weather station





D

weather computer



## 14 Fill in the Blank Text

2 / 2

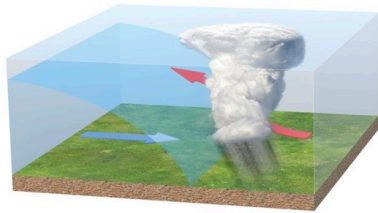
Fill in the blank.

Ilia has an assignment to make a circle graph showing the components of the atmosphere. He knows that the most abundant gas is nitrogen and that it makes up 78 percent of the atmosphere. The next most abundant gas is 1 oxygen ✓ , which makes up 21 percent of the atmosphere.

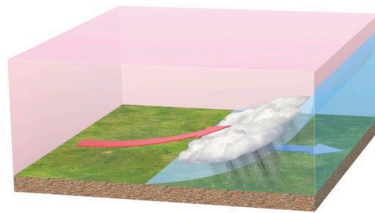
## 15 Multiple Choice

0 / 2

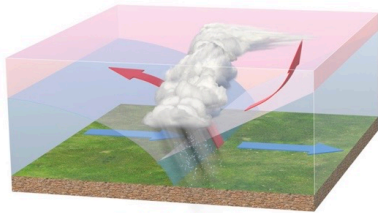
Choose the best answer.



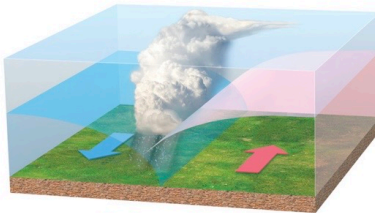
Cold Front



Warm Front



Occluded Front



Stationary Front

Which of the following statements is true for every type of weather front?

- ☐ when a front passes, warm and pleasant weather follows
- ☐ clouds can form when warm air masses interact with cool air masses
- ☒ the speed at which the air masses collide determines what type of precipitation falls
- ☐ cool air masses are forced up by warm air masses

✓

✗

16 Multiple Choice

2 / 2

Choose the best answer.

Mikhail watched the weather report on television. He saw the forecast for the coming week.

| Day of the week       | Sun   | Mon           | Tues   | Weds    | Thur    | Fri           | Sat     |
|-----------------------|-------|---------------|--------|---------|---------|---------------|---------|
| High temp.<br>(°C/°F) | 30/86 | 27.2/81       | 30/80  | 25.6/78 | 25.6/78 | 27.2/81       | 29.4/85 |
| Forecast              | Sunny | Partly cloudy | Cloudy | Rain    | Fog     | Partly cloudy | Sunny   |
| Relative humidity     | 29%   | 40%           | 90%    | 100%    | 100%    | 70%           | 30%     |

The meteorologist talked about how humidity affects weather and then predicted the humidity for the next week. Knowing that the amount of humidity would affect the coming weather, what does the amount of humidity on Tuesday indicate?

- ☐ There will be a low level of air pressure on that day.
- ☒ There will be a high level of water vapor in the air on that day.
- ☐ There will be a high level of air pressure on that day.
- ☐ There will be a low level of water vapor in the air on that day.



17 Ordering

2 / 2

**Ordering.**

Order the layers of the atmosphere in the correct order from the highest to lowest.

|   |              |   |
|---|--------------|---|
| 1 | thermosphere | ✓ |
| 2 | mesosphere   | ✓ |
| 3 | stratosphere | ✓ |
| 4 | troposphere  | ✓ |

**18 True/False****2 / 2****True or false.**

Water that condenses in the atmosphere near ground level is called dew point.

☐ True

☒ False

**Student's Correction**

Water that condenses in the atmosphere near ground level is called fog. Dew point is the point at which water vapor condenses.

**Correction Guidelines**

## 19 Fill in the Blank Drag and Drop

2 / 2

**Fill in the blanks.**

Unequal heating of the Earth's surface causes changes in 1 air pressure ✓. Warm air expands and becomes 2 less dense ✓. Cool,

3 dense air ✓ nearby pushes in underneath, causing the warmer air to rise. The movement of air parallel to Earth's surface is called

4 wind ✓.